

HEATING APPARATUS.

The heat in most heating apparatus is usually generated by an electrical module.

Examples of this apparatus are commonly used are:-

- i. Autoclave.
- ii. Hot air oven.
- iii. Incubator.
- iv. Water bath.
- v. Dryer.

Autoclave: The autoclave generates steam from water at a temperature above 100°C in a close chamber. At these temperature, the steam is above atmospheric pressure and the conditions are optimal for the sterilization of laboratory equipments, medical devices and media used in microbial culture. Bacteria can not survive in such environment, but viruses are not necessarily killed.

Hot-Air Oven: Hot-air oven are used mainly for drying laboratory equipments and surgical devices in dry air. Its design is such a way that internal circulation of dry-air occurs. Although some bacteria are killed at the higher temperature and some spores are likely survive.

Incubator: Incubator are used mainly for bacterial culture, but have additional uses within the laboratory. The incubator must maintain a constant temperature of $35 \pm 2^{\circ}\text{C}$.